

Sequential Trade Competitive Equilibrium

The same technological environment, just different trading arrangements: *trade occurs in every period*

Commodity space:

- Final goods (can be used for cons. or capital stock)
 - normalize price to 1
- Capital rental services
 - price r_t (measured in terms of time- t goods)
- Labor services
 - price w_t

Consumer's problem:

$$\text{Max}_{\{C_t, k_{t+1}, n_t, l_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(C_t, l_t)$$

$$\text{s.t. } l_t + n_t \leq 1 \quad l_t \geq 0 \quad n_t \geq 0$$

$$k_{t+1} \geq 0 \quad C_t \geq 0 \quad k_0 \text{ - given}$$

⊕ Budget constr. for every t :

$$C_t + k_{t+1} \leq r_t k_t + w_t n_t + \pi_t + (1-\delta) k_t$$

To solve it: (FOC) + (TVC) (like the DGM with linear technology)

Given $\{w_t, r_t, \pi_t\}_{t=0}^{\infty}$: find optimal $\{C_t, k_{t+1}, n_t, l_t\}_{t=0}^{\infty}$

Firm's Problem:

$$\pi_t = \text{Max}_{k_t, n_t} \{ F(k_t, n_t) - r_t k_t - w_t n_t \}, \quad \forall t \geq 0$$

$$\Rightarrow F_1(k_t, n_t) = r_t$$

$$F_2(k_t, n_t) = w_t$$

Given $\{w_t, r_t\}_{t=0}^{\infty}$, quantities $\{k_t, n_t, \pi_t\}_{t=0}^{\infty}$ solve the firm's problem.

A competitive equilibrium

consists of prices $\{r_t, w_t\}_{t=0}^{\infty}$ & quantities $\{G_t, k_{t+1}, n_t, l_t, \pi_t\}_{t=0}^{\infty}$ such that:

- 1) Given prices and dividends, consumers behave optimally
- 2) Given prices, firms behave optimally
- 3) prices are such that the markets clear
 - capital $\{k_t\}$: same in cons' & firms' problems
 - labor $\{n_t\}$: ——— " ———
 - goods: $G_t + k_{t+1} = F(k_t, n_t) + (1-\delta)k_t$

Remark: Since $\pi_t = F(k_t, n_t) - r_t k_t - w_t n_t$, the goods mkt clearing cond. is the same as the cons' budget constr. But this is not true, for example, if there are different types of consumers (e.g. only some of them own firms & receive dividends).

Welfare properties:

HW related: establish that the 1st & 2nd Welfare theorems hold.

General approach

The key step of setting up the recursive problems, as well as the recursive equilibrium, is choosing the set of state variables.

In contrast to the Social Planner's problem, the decision of agents in competitive equilibrium is affected by the market variables which the agents cannot control (e.g. prices).

These market variables:

- are endogenous (depend on the behaviour of the agents participating in the economy).
- may evolve over time (because individual agent's choices evolve over time)

⇒ the agents take into account how these aggregate variables evolve, the law of motion for these aggregate variables must be consistent with the agents' choices.

Let's try to be very general / abstract first:

In general, all the state variables can be classified into:

- 1) x - individual agent's state variables
(describe the agent-specific situation)
- 2) X - aggregate endogenous state variables
(describe the state of the markets that can be impacted by the choices of markets' participants)
- 3) Z - aggregate exogenous state variables (e.g. nature, etc.)
(affect the state of the market, not impacted by the agents' behavior).

Then the decision problem of an agent (consumer or firm) can be written down as:

$$V(x, X, Z) = \max_{y \in \Gamma(x, X, Z)} \{ F(x, X, Z, y) + \beta V(x', X', Z') \}.$$

s.t. 1) the law of motion for the individual state:

$$x' = h(x, X, Z, y)$$

↑
affected by individual choices

2) the law of motion for the aggregate state:

$$X' = H(X, Z)$$

- does not depend on individual state or choice
- agents take it as given

3) the law of motion for the nature's state:

$$Z' = M(Z)$$

- not affected by the agents' individual choices or aggregate outcomes

(so "nature" may be not the best interpretation - aka Global Warming!)

This problem says:

Given the current & the future state of the markets (X, X', Z, Z') , the agent with individual state x makes individual decision $y = g(x, X, Z)$.

In the equilibrium, the law of motion of the aggregate state variable must be consistent with the individual decisions of the agents.

Sometimes people separate individual and aggregate state variables by ; $V(x; X, z)$

Remarks :

1) In contrast to the sequential competitive equilibrium, recursive competitive equilibrium also describes out-of-equilibrium behavior because it describes the choices for all possible X , while only some values of X may obtain in equilibrium.

2) Choosing X is not trivial...

First, X should include all the aggregate variables that determine prices (or prices themselves). Second, X should be sufficient, along with z , for describing the evolution of X .

For example, in the growth model we've considered, prices r and w are determined by (K, N) . But aggregate (K, N) may be not sufficient to describe the evolution of prices when agents are heterogeneous (then the distribution of k across people, not just total/average K , matters!)

Recursive Competitive Equilibrium (RCE) in a neoclassical growth model

- The same environment as before:

- consumers • own capital & shares in firms
 - decide how much to work / rest
 - decide how much to consume / save
- firms • choose inputs to max. profit
 - pay dividends to shareholders

- Trade occurs in every period:

(recursive setup $\Rightarrow \exists$ today and the future $\Rightarrow$
 $\Rightarrow$ cannot time the trade differently)

- State space:

- no nature variable ($\nexists Z$)

(other settings: technology $Z = F(K, N)$, Z evolves over time, deterministically or stochastically)

- aggregate state variables.

what does consumer take as given? prices & dividends
 $\Rightarrow$ obvious candidate is

$$X = (r, w, \pi)$$

need to think about the aggregate law of motion $X' = H(X) \dots$

This would be correct...

But the state space can be reduced. !!!

(always try to do it if possible, because computation time increases exponentially in the # of state variables).

Denote by k, n : individual choices (small letters)
 K, N : aggregate variables (capital letters)

Running ahead, we foresee that

$$r = F_1(K, N) = \tilde{r}(K, N)$$

$$w = F_2(K, N) = \tilde{w}(K, N)$$

$$\begin{aligned}\pi &= F(K, N) - rK - wN = F(K, N) - \tilde{r}(K, N) \cdot K - \tilde{w}(K, N) \cdot N \\ &= \tilde{\pi}(K, N)\end{aligned}$$

Thus, knowing aggregate K, N is sufficient for recovering (r, w, π)

$\Rightarrow X = (K, N)$ is a natural candidate.

Can we reduce the state space even further?

For an individual, e/n choice is determined by k and aggregate state variables.

$\Rightarrow$ Aggregate labor is a function of aggregate state (at least in the representative agent economy).

So it makes sense to "guess" that we can only have one state variable, K ,
aggr.

with N being a fn of it, and this function must be consistent with individual labor choices.

- individual state, k (individual capital stock)

Consumer problem:

Consumer problem :

$$V(k, K) = \max_{\substack{0 \leq k' \\ 0 \leq l \leq 1}} \{ u(c, l) + \beta V(k', K') \}$$

$$\text{s.t. 1) } c + k' \leq w(K)(1-l) + (1+r(K)-\delta)k + \pi(K)$$

(CP)

2) $r(K), w(K), \pi(K)$ are exogenous
(for the consumer)

$$3) K' = H(K)$$

the law of motion for the aggregate state

$$\begin{aligned} \Rightarrow k' &= g^k(k, K) \\ c &= g^c(k, K) \\ n=1-l &= g^n(k, K) \end{aligned} \quad \left. \vphantom{\begin{aligned} \Rightarrow k' &= g^k(k, K) \\ c &= g^c(k, K) \\ n=1-l &= g^n(k, K) \end{aligned}} \right\} \text{policy functions}$$

Firm's problem :

(FP) $\text{Max}_{k, n} F(k, n) - r(K) \cdot k - w(K) \cdot n$

$$\Rightarrow k^*(K), n^*(K), \pi(K)$$

The Recursive Competitive Equilibrium consists of:

- price functions : $r(K), w(K)$
- policy functions : $\left. \begin{array}{l} g^k(k, K) \\ g^c(k, K) \\ g^n(k, K) \end{array} \right\} \text{consumers}$
(quantities)

$$\left. \begin{array}{l} \ddot{g}^n(k, K) \\ k^f(K) \\ n^f(K) \\ \pi(K) \end{array} \right\} \text{firms}$$

- value function: $V(k, K)$

- the law of motion for aggregate state $H(K)$

such that:

1) Given $r(K), w(K), \pi(K)$ and $H(K)$,

$V(k, K)$ solves (CP)

$g^i(k, K)$ - corresponding policies

2) Given $r(K), w(K)$,

$k^f(K), n^f(K)$ and $\pi(K)$ solve (FP)

3) Markets char:

capital: $k^f(K) = K$, $\forall K$

$\uparrow$
representative firm

note: in equilibrium, $K = k$ (capital supply)

but k, K are state var-s for the cons.,

the problem is solved for $\forall$ combinations of (k, K) , not just $k = K$.

But we will use $k = K$ below.

labor: $n^f(K) = N = g^e(k, K)$, $\forall K$

$\uparrow$
representative firm

$\uparrow$
representative consumer

aggregate labor

$k = K$ in equilibr.

aggregate labor

goods: $C + K' = Y + (1-d)K$
(in aggregate variables)

$$\Leftrightarrow g^c(K, K) + H(K) = F(K, n^t(K)) + (1-d)K, \quad \forall K$$

4) Consistency (of aggr. law of motion with ind. decisions)

$$H(K) \underset{\uparrow}{=} g^{k'}(K, K), \quad \forall K$$

representative consumer

Welfare properties of the RCE:

Social Planner's allocation:

$$V(K) = \max_{0 \leq K' \leq F(K,1) + (1-\sigma)K} U(F(K,1) + (1-\sigma)K - K') + \beta V(K')$$

$\Leftrightarrow K' = g(K)$ is such that (EE^*)

$$u'(F(K,1) + (1-\sigma)K - g(K)) = \beta (F_1(g(K),1) + (1-\sigma)g(K)) \cdot u'(F(g(K),1) + (1-\sigma)g(K) - g(g(K)))$$

1st Welfare theorem:

Competitive equilibrium is efficient.

need to show that $K' = H(K)$ solves the (SP) problem, namely that $g(K) = H(K)$ solves (EE^*) .

(CP) in RCE:

$$V(k,K) = \max_{K' \leq (1+r(K)-\sigma)k + \pi(K)} U(\underbrace{(1+r(K)-\sigma)k + \pi(K) - K'}_{g^c(k,K)}) + \beta V(K', H(K))$$

(FOC): $u'(c) = \beta V_1(K', H(K))$

(EC): $V_1(k,K) = u'(c) \cdot (1+r(K)-\sigma)$

$\Rightarrow (EE): u'(g^c(k,K)) = \beta (1+r(K)-\sigma) \cdot u'(g^c(g^k(k,K), K'))$
 s.t. $K' = H(K)$ $\forall k, K$

In particular, it must hold for $k=K$:

$$u'(g^c(K,K)) = \beta (1+r(H(K))-\sigma) \cdot u'(g^c(g^k(K,K), K')) \quad K' = H(K)$$

$$= F(K,1) + (1-\sigma)K - H(K) \quad = F(H(K),1) + (1-\sigma)H(K) \quad = H(K) \quad = H(K)$$

$$\begin{aligned}
 &= F(k, 1) + (1-\delta)k - H(k) &= F_1(H(k), 1) &= H(k) &= H(k) \\
 &\text{(goods mkt clearing)} &\text{(FP)} &\text{(consistency)}
 \end{aligned}$$

$$= F(H(k), 1) + (1-\delta)H(k) - H(H(k))$$

$$\Rightarrow g(k) = H(k) \text{ solves } (EE^*) \quad \square$$

2nd Welfare theorem:

Pareto Optimal allocation can be achieved in a competitive equilibrium

Proof: trivial:

$$\text{define } r(k) = F_1(k, 1)$$

$$w(k) = F_2(k, 1)$$

$$\pi(k) = F(k, 1) - r(k) \cdot k - w(k) \cdot 1$$

$$H(k) = g(k) \text{ (optimal policy in (SP))}$$

$\Rightarrow$ all equilibrium conditions are immediately verified $\square$